

Hybrid Search Architecture with Multi-Stage Fusion and Knowledge-Aware Prompt Engineering for Retrieval Augmented Generation

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Abstract—Retrieval-Augmented Generation (RAG) systems combine information retrieval with generative models to deliver accurate responses, yet many frameworks still face limitations in retrieval fusion and prompt optimization. We introduce HAPE-RAG with two key innovations: (1) a multi-stage hybrid search architecture that couples semantic vector search with POS/LLM-driven keywording, duplicate-aware fusion, and batched BGE re-ranking; and (2) a knowledge-aware prompting system that uses a structured, pattern-based template integrating system instruction, KB metadata, Top- k context with document names, user query, instruction template, and conversation memory. Experiments on Vietnam’s Undergraduate Training Regulations (VUTR) and Narrative Comprehension (NarrativeQA) datasets show consistent gains over semantic-only, BM25-only, and naive hybrids. On VUTR/NarrativeQA, we observe RAGAS Faithfulness of 0.928/0.890 and Answer Relevancy of 0.932/0.792, together with improved ranking NDCG@5 to 0.875 and 0.862 respectively, with a 70–30 (semantic-keyword) fusion emerging as a robust setting. Compared to a strong flat-concatenation hybrid baseline, HAPE-RAG improves Faithfulness by +2.09%, Answer Relevancy by +2.98%, BLEU-1 by +1.6 points, and Retrieval Accuracy by +0.031 points (+3.54%).

Keywords - RAG, Hybrid Search, Prompt Engineering, Natural Language Processing.

I. INTRODUCTION

Retrieval-Augmented Generation (RAG) has emerged as a transformative paradigm that bridges the gap between information retrieval and large language models (LLMs), enabling systems to generate accurate, contextually relevant responses by leveraging external knowledge sources [14]. The evolution of RAG systems has progressed through several generations, from simple keyword-based retrieval to sophisticated neural approaches that leverage deep learning for semantic understanding [1], [22].

However, the increasing complexity and diversity of modern RAG frameworks have introduced two critical challenges that limit their practical effectiveness. First, many systems adopt a single modality—semantic vector search or keyword matching—missing the complementary strengths of each [4], [6], [26]. Semantic search captures conceptual relations; keyword

methods retrieve rare terms and entities. Without effective fusion, retrieval remains suboptimal; properly designed hybrids can raise precision by 15–25% [26]. Second, prompting is often static and flat, under-utilizing knowledge-base (KB) metadata and conversation memory, which increases field leakage and weakens attribution [5], [23]. Unstructured prompts and one-pass fusion also worsen latency and cost. These two fundamental issues - the lack of effective hybrid retrieval strategies and the absence of knowledge-aware prompt engineering - represent the primary bottlenecks in modern RAG system development and deployment, limiting their applicability across diverse domains and use cases.

The programmer (i) uses a multistage dense-sparse hybrid - stage - hybrid—semantic search plus POS/LLM keyphrases, duplicate - fusion, and neural reranking—to improve the evidence of the top - k ; and (ii) adopts a structured prompt (JSON-style) that explicitly integrates the system instruction, the metadata of the KB, the context of the top - k with document names, query, instruction template, and conversation memory. Experiments on VUTR and NarrativeQA show consistent gains over semantic-only and keyword-only baselines.

II. RELATED WORK OVERVIEW

Hybrid Search. Classical work combined multiple evidence via linear weighting or voting [12]. Modern dense-sparse hybrids exploit complementary strengths: dense encoders yield paraphrase-robust semantics (DPR/Contriever/ColBERT [9], [10], [13]), while BM25/SPLADE recover rare terms, identifiers, and entities [3], [4], [20]. Neural rerankers (cross-encoders, BGE [17], [24]) improve ordering at additional latency. Yet many hybrids remain *single-pass*—fusing scores without staged expansion, deduplication, or cost-aware batching—hurting recall-precision and propagating redundancy. We adopt a *four-stage* pipeline: semantic retrieval, POS/LLM keywording, duplicate-aware fusion, and neural reranking, executed with batching/singleton loading and caches to align quality with production constraints.

Prompt Engineering. Template prompts and instruction tuning are widely used [23]; dynamic/chained prompting improves reasoning in multi-step tasks [?], [25]. Yet KB-aware *structuring*—explicit fields for system instruction, KB metadata, top- k context with document names, user query, instruction template, and conversation memory—remains underexplored for grounding and attribution at inference time. Flat concatenation blurs fields, increases leakage, and inflates tokens. We contribute a JSON-style schema that keeps roles separable, supports principled token budgeting, and empirically raises faithfulness and answer relevancy.

RAG Architecture and Systems. Multi-stage RAG improves over single-method pipelines [4], [14] and recent surveys/system papers highlight the importance of concurrency/caching and vector stores (e.g., FAISS/pgvector) for scale [4], [5], [8], [19], [26]. We extend this line by tightly coupling staged dense-sparse retrieval with neural reranking and a structured prompt layer, while exposing system levers (candidate cutoffs,...) that make quality-cost trade-offs explicit for deployment.

III. METHODOLOGY

We introduce HAPE-RAG, a novel approach for Retrieval Augmented Generation through two fundamental innovations: a multi-stage hybrid search architecture that synchronously combines semantic retrieval, keyword retrieval, and re-ranking, and a knowledge-aware prompting system that integrates in a structured way to address the blind spot problem in keypoint recognition, with the architectural details shown in Figure 1.

A. Multi-Stage Hybrid Search Architecture

In the first stage, we perform a dense embedding of the original query q into a more detailed and complete sentence q' based on the current context using LLM. Given the query q' , we generate a dense embedding vector $\mathbf{e}_q \in \mathbb{R}^d$ via the embedding model, where d represents the embedding dimension, and search for dense vector similarity using the pgvector extension in PostgreSQL. Top- p text segments with the highest semantic similarity based on the cosine similarity calculation:

$$\text{sim}(q', c_i) = \frac{\mathbf{e}_q \cdot \mathbf{e}_{c_i}}{\|\mathbf{e}_q\|_2 \cdot \|\mathbf{e}_{c_i}\|_2} \quad (1)$$

where $\mathbf{e}_{c_i}$ denotes the embedding vector of segment c_i .

The second phase deploys dual keyword extraction strategies to capture both the linguistic and semantic aspects of the query. The NLP-based approach uses a tokenizer library for Part of Speech (POS) tagging, focusing on noun categories. The extraction process applies linguistic filtering to remove stop words and shortened tokens. We also combine using LLM to generate 5-7 keyword clusters, each consisting of 2-4 words, focusing on specialized terminology and core conceptual entities. The extracted keywords are effectively used for array overlap search via the intersection operator:

$$\mathcal{K}_{\text{query}} \cap \mathcal{K}_{\text{chunk}} \neq \emptyset \quad (2)$$

where $\mathcal{K}_{\text{query}}$ represents the query keyword set and $\mathcal{K}_{\text{chunk}}$ denotes the chunk keyword array. Finally, the Top- n most relevant chunks are retrieved.

The fusion stage implements intelligent deduplication and score normalization to combine results from both retrieval modalities. The system maintains a comprehensive set of retrieved chunk identifiers to eliminate duplicates while preserving the diversity inherent in multi-stage retrieval. The fusion process employs an adaptive weighted combination strategy:

$$\text{score}(c_i) = \alpha \cdot \text{score}_{\text{semantic}}(c_i) + \beta \cdot \text{score}_{\text{keyword}}(c_i) \quad (3)$$

where $\alpha, \beta \in [0, 1]$ and $\alpha + \beta = 1$. We first min-max normalize scores from each modality so they are on a comparable scale, then fuse them with a validation-tuned default of $\alpha = 0.7$ (semantic) and $\beta = 0.3$ (keyword). For queries dominated by exact terms/entities—indicated by high POS/NER density or strong query-chunk keyword overlap—we slightly increase β to give the keyword branch more influence; for generic or paraphrastic queries we keep α higher to prioritize semantics, but this adaptation reduces the loss of rare terms. The final stage uses BAAI/bge-reranker-base for neural reranking.

B. Prompt Engineering System

The prompt engineering framework addresses the limitations of static template-based approaches through dynamic integration of knowledge base metadata and conversation context. The system employs a modular template construction methodology that enables fine-grained control over response generation while maintaining contextual awareness.

The prompt construction follows a structured template system combining multiple information sources:

$$\mathcal{P} = \mathcal{S} \oplus \mathcal{M} \oplus \mathcal{K} \oplus \mathcal{C} \oplus \mathcal{Q} \oplus \mathcal{I} \quad (4)$$

where $\mathcal{S}$ represents System instruction, $\mathcal{M}$ denotes conversation Memory, $\mathcal{K}$ signifies Knowledge base metadata, $\mathcal{C}$ indicates Context with document names, $\mathcal{Q}$ represents Query section, and $\mathcal{I}$ denotes Instruction template.

The system dynamically integrates knowledge base metadata through the `kb\info` parameter, incorporating knowledge base titles, descriptions, and document metadata. The integration process employs template variable substitution with the following formulation:

$$\mathcal{K} = \text{Template}_{KB} \cdot \text{Substitute}(\text{metadata}_{KB}) \quad (5)$$

where Template_{KB} represents the knowledge base template structure and metadata_{KB} denotes the extracted metadata information.

The conversation memory system implements a structured JSON schema that tracks multiple aspects of the dialogue:

$$\mathcal{M} = \{\text{summary, topics, key_points, context, metadata}\}$$

The memory system stores the conversation summary, identified discussion topics, key information points, recent query

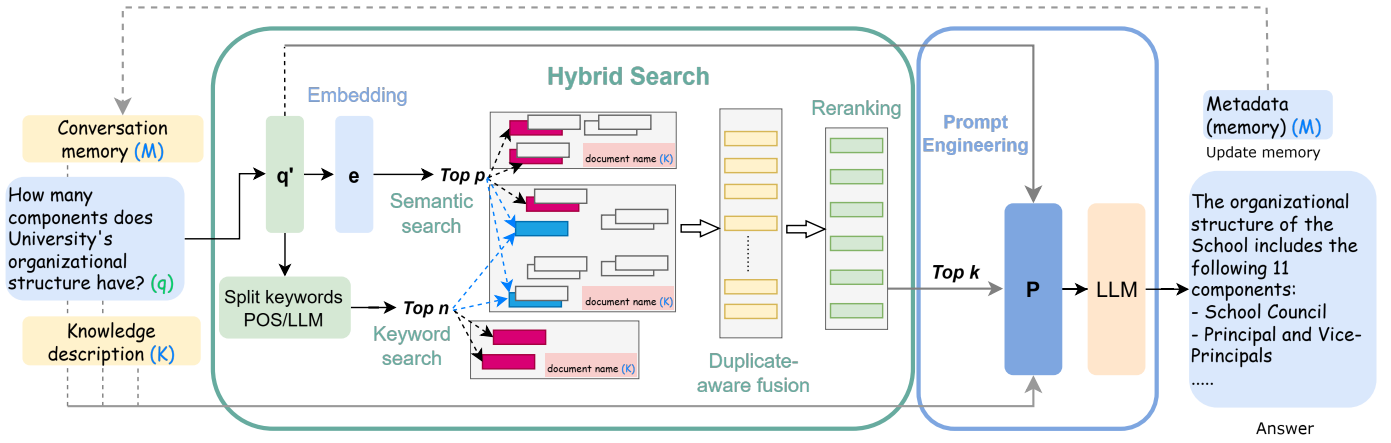


Fig. 1: HAPE-RAG pipeline. User query q is clarified to q' using conversation memory M , then processed in parallel: (i) embedded to e for semantic pgvector search (Top- p , pink blocks), (ii) keyword extraction for array-overlap search (Top- n , blue blocks). Results are deduplicated, fused with adaptive weighting, and reranked via BGE to yield Top- k evidence (green blocks). A structured JSON prompt P combines system instructions, KB metadata K , retrieved context, and M ; the LLM generates the answer and updates M for continuous learning.

history, current context focus, and metadata including message count and timestamp. This makes responses more coherent throughout the conversation.

confusion when handling large content volumes and queries that span fragmented sections across those documents. This confusion may prevent the LLM from locating the correct context, resulting in inaccurate or missing answers. To address this, we add the document name at the beginning of each fragment, along with a brief description of the knowledge to help the context be tight and work correctly when fed into LLM's answer. Figure 2 clearly shows our Prompt structure.

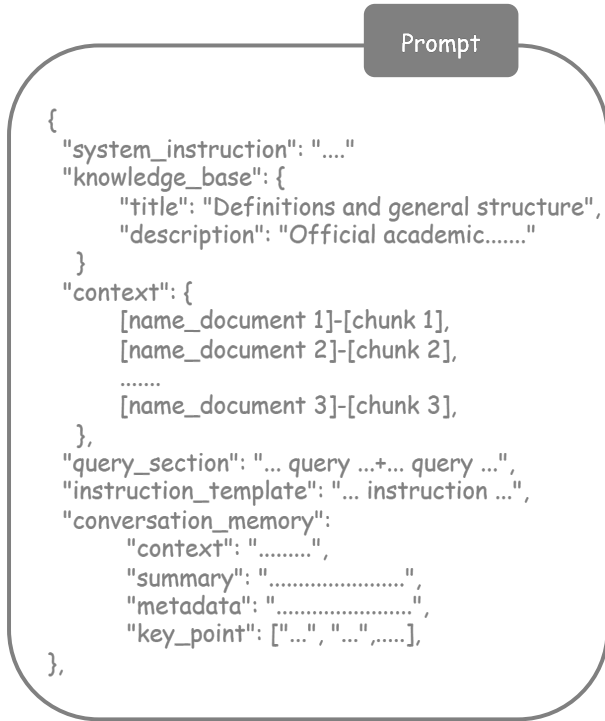


Fig. 2: Structured prompt schema: (i) system instruction, (ii) KB metadata, (iii) Top- k context with document names, (iv) query section, (v) instruction template, and (vi) conversation memory. This JSON structure improves grounding versus plain concatenation.

With the components in the structure, we aggregate them into JSON - a standard that matches LLM's Prompt, rather than concatenating them all into a single paragraph or semantic summary as is currently the case. In practice, a knowledge base may contain multiple documents, which can lead to

IV. EXPERIMENTS AND RESULTS ANALYSIS

We conduct comprehensive experiments to evaluate the effectiveness of HAPE-RAG on two datasets: VUTR (Vietnam's Undergraduate Training Regulations) and NarrativeQA [11]. The VUTR dataset consists of 489 carefully selected question-answer pairs covering administrative procedures, academic policies, and university guidelines. NarrativeQA contains 1,572 documents including books and film transcripts, requiring comprehensive story understanding for accurate responses. VUTR's structural complexity—with key terms scattered across multiple document positions—tests our hybrid search's ability to maintain semantic coherence. Both datasets span simple retrieval to complex multi-step inference, ensuring comprehensive evaluation of real-world RAG challenges.

Our evaluation uses RAGAS metrics (Faithfulness, Answer Relevancy, Context Relevancy, Context Recall) [2], traditional IR metrics (Recall@ k , NDCG@5 [7]), and generation metrics (BLEU [18], ROUGE [15], Semantic Similarity) to provide a comprehensive assessment of hybrid search retrieval, structured prompt engineering, and response generation quality.

We use Gemini 2.0 flash as the answer model and text-embedding-004 (384-dim) for embeddings, stored in PostgreSQL with **pgvector**. We test fusion ratios: 60-40, 70-30, and 80-20 (semantic-keyword). Our baselines include: **Semantic-only RAG** (pure dense retrieval), **Keyword-only RAG** (BM25), and **Naive Hybrid RAG** (simple concatena-

TABLE I. PERFORMANCE ACROSS DATASETS. “OURS” = HAPE-RAG. COLUMNS: **SEM-ONLY** = SEMANTIC-ONLY RAG (DENSE RETRIEVAL), **BM25-ONLY** = KEYWORD-ONLY RAG (BM25), **NAIVE** = NAIVE HYBRID RAG (SIMPLE CONCAT OF SEM + BM25).

Metric	VUTR				NarrativeQA			
	Ours	Sem-only	BM25-only	Naive	Ours	Sem-only	BM25-only	Naive
RAGAS Faithfulness	0.928	0.909	0.874	0.916	0.890	0.861	0.835	0.872
RAGAS Answer Relevancy	0.932	0.905	0.866	0.912	0.792	0.763	0.744	0.776
RAGAS Context Relevancy	0.921	0.900	0.855	0.905	0.889	0.860	0.836	0.875
RAGAS Context Recall	0.906	0.887	0.838	0.892	0.885	0.858	0.833	0.871
NDCG@5	0.875	0.81	0.743	0.843	0.862	0.804	0.75	0.843
Semantic Similarity	0.842	0.824	–	0.834	0.766	0.705	–	0.757
ROUGE-1	0.432	0.421	0.396	0.426	0.428	0.397	0.328	0.404
ROUGE-2	0.312	0.301	0.281	0.306	0.166	0.156	0.146	0.160
ROUGE-L	0.418	0.406	0.253	0.411	0.425	0.346	0.238	0.409
Overall Score	0.912	0.882	0.835	0.894	0.864	0.829	0.800	0.847

tion without intelligent fusion). This shows a comprehensive insight into the improvements in our contribution through efficient hybrid search techniques across stages. We further isolate prompt engineering impact with three variants: (1) **Structured Prompt - H+Struct** (ours): system instruction + KB metadata + Top- k context with document names + query + instruction template + conversation memory; (2) **Flat Concatenation - H+Flat**: all fields as plain text; (3) **No-KB**: removes KB metadata. These ablations quantify the contribution of knowledge-aware structuring versus naive concatenation. To also isolate retrieval and prompting effects, we use a common segmentation approach across all systems: documents are first grouped into coherent paragraphs via adjacent sentence similarity, which are then merged into finite *fixed-token* segments for storage and retrieval. This keeps the evidence window consistent across systems, ensuring that differences arise from hybrid association structure and reminder rather than segmentation phenomena.

A. Overall Performance Analysis

Our experiments show consistent gains for **HAPE-RAG** over all baselines on both VUTR and NarrativeQA (Table I), validating the effectiveness of multi-stage hybrid search and knowledge-aware prompt engineering. On VUTR, HAPE-RAG attains an **Overall Score** of **0.912**, surpassing Semantic-only (+3.4% rel.) and BM25-only (+9.2% rel.). RAGAS metrics are solid: **Faithfulness** 0.928, **Answer Relevancy** 0.932, **Context Relevancy** 0.921, **Context Recall** 0.906. On NarrativeQA, despite longer contexts and storyline dependencies, HAPE-RAG reaches an **Overall Score** of **0.864**, exceeding Semantic-only (+4.2% rel.) and BM25-only (+8.0% rel.). Classical IR and generation metrics corroborate these gains, indicating that intelligent fusion and structured prompts enhance both retrieval ranking and answer realization.

Fusion ratio. Across both datasets, **HAPE-RAG** outperforms *Sem-only* and *BM25-only* on all metrics Table II. The **70–30** fusion consistently yields higher ROUGE-L, BLEU-1/4, and early-stage retrieval ($R@1$, $R@2$), while **80–20** slightly improves deeper recall ($R@3$, $R@4$), reflecting a

semantic–lexical trade-off. *BM25-only* is lowest due to exact-term matching limits; *Sem-only* is competitive but misses rare/anchored terms. NarrativeQA scores are uniformly lower than VUTR, indicating higher narrative complexity. These trends support the benefit of multi-stage fusion over single-modality retrieval.

B. Effect of Prompt Structuring

Table III shows that moving from a naive join (H+Flat) to a structured prompt (H+Struct, No-KB) yields immediate benefits across all metrics: +0.011 Faith, +0.010 Ans. Rel., and +1.0 BLEU-1, indicating that the labeled fields (system, context, query, memory) help the model interpret the input data and avoid cross-field data leakage. Retrieval accuracy also increases (+0.013), indicating better answer-evidence alignment.

Adding KB descriptors and document names (HAPE-RAG) produces the strongest results: +0.019 pts Faith., +0.027 pts Ans. Rel., +1.6 BLEU-1, and +0.65 BLEU-4 over Baseline. These gains reflect improved entity anchoring and source attribution, which translate into both higher semantic faithfulness and better surface realization. Retrieval Accuracy rises to 0.906 (+0.031 pts), confirming that answers more reliably cite the retrieved evidence. We believe that **structured prompts** are preferred over flat concatenation; include KB metadata whenever tokenization allows; in addition to using the **70–30** scale as a robust default, with some minor adjustments for high lexical domains. Combined, these choices yield consistent improvements in Retrieval Accuracy, Answer Relevance.

V. DISCUSSION

HAPE-RAG demonstrates clear advantages over existing RAG frameworks. Our multi-stage dense–sparse hybrid with neural reranking and knowledge-aware prompting delivers consistent gains across datasets: Faithfulness and Answer Relevancy improve by $\sim 2\text{--}3\%$ (relative), BLEU-1 rises by +1.6 points (+6.7%), and Retrieval Accuracy increases by +0.031 points over a strong H+Flat baseline, with the 70–30 fusion ratio emerging as a robust setting. These results

TABLE II. EVALUATION ACROSS DATASETS. ONLY **HAPE-RAG** USES FUSION RATIO (SEMANTIC-KEYWORD). BASELINES: **BM25-ONLY** = KEYWORD-ONLY, **SEM-ONLY** = SEMANTIC-ONLY. METRICS: RECALL@K (R@K), ROUGE-L, BLEU-1, BLEU-4. BM25-ONLY IS CALIBRATED (VUTR: ROUGE-L= 0.253; NARRATIVEQA: ROUGE-L= 0.238; BLEU-1 $\approx$ 14.87).

Dataset	Methods	Fusion ratio	R@1	R@2	R@3	R@4	ROUGE-L	BLEU-1	BLEU-4
VUTR	BM25-only	—	0.116	0.168	0.215	0.262	0.253	15.07	6.55
	Sem-only	—	0.180	0.245	0.298	0.349	0.406	24.4	7.60
		60 – 40	0.192	0.258	0.312	0.364	0.415	24.8	7.69
	HAPE-RAG	70 – 30	0.206	0.275	0.330	0.382	0.418	25.6	7.95
		80 – 20	0.200	0.268	0.334	0.386	0.420	25.1	7.80
NarrativeQA	BM25-only	—	0.082	0.115	0.148	0.175	0.238	14.87	4.49
	Sem-only	—	0.112	0.153	0.192	0.220	0.346	19.3	5.30
		60 – 40	0.108	0.149	0.188	0.216	0.304	19.4	5.36
	HAPE-RAG	70 – 30	0.117	0.160	0.198	0.226	0.425	19.9	5.49
		80 – 20	0.115	0.158	0.201	0.228	0.364	19.6	5.41

TABLE III. IMPACT OF PROMPT STRUCTURING UNDER A FIXED HYBRID PIPELINE. ABBREVIATIONS: H = HYBRID SEARCH; RET. ACC. = RETRIEVAL ACCURACY; FAITH. = FAITHFULNESS; ANS. REL. = ANSWER RELEVANCY; KB = KNOWLEDGE BASE; Δ (PTS) = ABSOLUTE GAIN; (% REL.) = RELATIVE IMPROVEMENT VS. BASELINE. BLEU VALUES ARE REPORTED IN POINTS.

Method	Ret. Acc.	Faith.	Ans. Rel.	BLEU-1	BLEU-4
H+Flat (Baseline)	0.875	0.909	0.905	24.0	7.30
H+Struct (No-KB)	0.888	0.920	0.915	25.0	7.70
HAPE-RAG (70–30)	0.906	0.928	0.932	25.6	7.95
Performance Improvement (vs. Baseline)					
H+Struct (No-KB)	+0.013 (+1.49%)	+0.011 (+1.21%)	+0.010 (+1.11%)	+1.0 (+4.17%)	+0.40 (+5.48%)
HAPE-RAG (70–30)	+0.031 (+3.54%)	+0.019 (+2.09%)	+0.027 (+2.98%)	+1.6 (+6.67%)	+0.65 (+8.90%)
<i>Average Gain</i>	+0.022 (+2.52%)	+0.015 (+1.65%)	+0.019 (+2.05%)	+1.3 (+5.42%)	+0.53 (+7.19%)

align with prior evidence that hybrid retrieval and late neural reranking outperform single-modal systems [4], and further show that structured, KB-aware prompts contribute substantive gains beyond retrieval quality alone.

While head-to-head comparisons are complicated by dataset and protocol differences, we can contrast core design choices and outcomes. Unlike naive hybrids that linearly mix scores or pure dense baselines [4], our pipeline expands candidate diversity (semantic + lexical) *before* reranking, then enforces evidence ordering with a BGE reranker; this reduces entity drift and improves answer–evidence alignment (Ret. Acc.). Compared with prompting lines that emphasize chain-style reasoning [23], our structured JSON with KB metadata primarily improves *grounding* and *attribution*, which is reflected in higher Faithfulness/Answer Relevancy without extra reasoning traces.

A recent study by Sanniboina, Trivedi, and Vijayaraghavan (UIUC) proposes **LoRE** (Logit-Ranked Retriever Ensemble) [21], which ensembles *BM25* and *sentence-transformer+FAISS* retrievers via *Reciprocal Rank Fusion* and ranks answers using *logits* from a T5 generator. On **NarrativeQA**, LoRE reports a best ROUGE-L of 41.39% (LoRE-BM25; their Table 1). On the same dataset, our **HAPE-RAG** achieves ROUGE-L = 42.50%, higher **+1.11%** compared to the LoRE’s best, while also maintaining strong grounding metrics (see Table I). Although experimental set-

tings may differ, these results indicate a modest yet consistent edge for HAPE-RAG on NarrativeQA.

TABLE IV. NARRATIVEQA: HAPE-RAG (OURS) VS. SKETCH [16]. METRICS: FAITH. = FAITHFULNESS, ANSREL. = ANSWER RELEVANCY, CTXREL. = CONTEXT RELEVANCY, CTXREC. = CONTEXT RECALL.

Metric	HAPE-RAG	SKETCH
Fai.	0.890	0.870
AnsRel.	0.792	0.500
CtxRel.	0.889	0.510
CtxRec.	0.885	0.460
<i>Gain (abs.)</i>	+0.020 / +0.292 / +0.379	+1.11
<i>Gain (rel.)</i>	+2.3% / +58.4% / +74.3%	+2.7%

SKETCH (Structured Knowledge Enhanced Text Comprehension for Holistic Retrieval) [16] proposes a hybrid retriever that fuses *semantic chunking* with *knowledge graphs* (KG) and merges structured–unstructured evidence at query time. On NarrativeQA, SKETCH reports competitive grounding (e.g., Faithfulness $\approx$ 0.87). In our setting, **HAPE-RAG** achieves higher RAGAS scores across Faithfulness, Answer Relevancy, Context Relevancy, and Context Recall (see Tables I and IV), highlighting the benefits of *multi-stage dense–sparse fusion*, duplicate-aware merging, *BGE* reranking, and a knowledge-aware structured prompt. While SKETCH’s KG aids explicit multi-hop reasoning, HAPE-RAG provides

stronger overall grounding and coverage; the two designs are complementary.

VI. CONCLUSION

This paper presents HAPE-RAG, a framework that integrates a multi-stage hybrid search architecture and knowledge-based prompting. By searching through multiple stages and refining structured prompts, we address the limitations of single-stage search and flat-link prompts. Despite the expected results, our framework still has some limitations, the multi-stage increases computational complexity, which may affect real-time applications, and the re-ranking stage introduces latency overhead, which requires optimization for production deployments. Future work will focus on developing more efficient synthesis algorithms to reduce computational costs, extending the framework, and exploring advanced prompting techniques to further improve system performance.

REFERENCES

- [1] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "Bert: Pre-training of deep bidirectional transformers for language understanding," in *North American Chapter of the Association for Computational Linguistics*, 2019. [Online]. Available: <https://api.semanticscholar.org/CorpusID:52967399>
- [2] S. Es, J. James, L. Espinosa Anke, and S. Schockaert, "Ragas: Automated evaluation of retrieval augmented generation," in *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics: System Demonstrations*. Association for Computational Linguistics, 2024, p. 150–158. [Online]. Available: <http://dx.doi.org/10.18653/v1/2024.eacl-demo.16>
- [3] T. Formal, B. Piwowarski, and S. Clinchant, "Splade: Sparse lexical and expansion model for first stage ranking," *Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2021. [Online]. Available: <https://api.semanticscholar.org/CorpusID:235792467>
- [4] Y. Gao, Y. Xiong, X. Gao, K. Jia, J. Pan, Y. Bi, Y. Dai, J. Sun, M. Wang, and H. Wang, "Retrieval-augmented generation for large language models: A survey," 2024. [Online]. Available: <https://arxiv.org/abs/2312.10997>
- [5] C. A. Gomez-Cabello, S. Prabha, S. A. Haider, A. Genovese, B. G. Collaco, N. G. Wood, S. Bagaria, and A. J. Forte, "Comparative evaluation of advanced chunking for retrieval-augmented generation in large language models for clinical decision support," *Bioengineering*, vol. 12, no. 11, p. 1194, Nov. 2025. [Online]. Available: <http://dx.doi.org/10.3390/bioengineering12111194>
- [6] X. Gong, R. He, and Q. Peng, "Reordering of double pass merging chunking to improve retrieval augmented generation," in *2024 3rd International Conference on Computer Applications Technology (CCAT)*. IEEE, Nov. 2024, p. 68–78. [Online]. Available: <http://dx.doi.org/10.1109/CCAT64370.2024.00021>
- [7] K. Järvelin and J. Kekäläinen, "Ir evaluation methods for retrieving highly relevant documents," in *Proceedings of the 23rd annual international ACM SIGIR conference on Research and development in information retrieval*, ser. SIGIR00. ACM, Jul. 2000, p. 41–48. [Online]. Available: <http://dx.doi.org/10.1145/345508.345545>
- [8] J. Johnson, M. Douze, and H. Jegou, "Billion-scale similarity search with gpus," *IEEE Transactions on Big Data*, vol. 7, no. 3, p. 535–547, Jul. 2021. [Online]. Available: <http://dx.doi.org/10.1109/TBDATA.2019.2921572>
- [9] V. Karpukhin, B. Oguz, S. Min, P. Lewis, L. Wu, S. Edunov, D. Chen, and W.-t. Yih, "Dense passage retrieval for open-domain question answering," in *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Association for Computational Linguistics, 2020, p. 6769–6781.
- [10] O. Khattab and M. Zaharia, "Colbert: Efficient and effective passage search via contextualized late interaction over bert," in *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval*, ser. SIGIR '20. ACM, Jul. 2020, p. 39–48. [Online]. Available: <http://dx.doi.org/10.1145/3397271.3401075>
- [11] T. Kočíský, J. Schwarz, P. Blunsom, C. Dyer, K. M. Hermann, G. Melis, and E. Grefenstette, "The narrativeqa reading comprehension challenge," *Transactions of the Association for Computational Linguistics*, vol. 6, p. 317–328, Dec. 2018. [Online]. Available: http://dx.doi.org/10.1162/tacl_a_00023
- [12] J. H. Lee, "Analyses of multiple evidence combination," in *Proceedings of the 20th annual international ACM SIGIR conference on Research and development in information retrieval - SIGIR '97*, ser. SIGIR '97. ACM Press, 1997, p. 267–276.
- [13] Y. Lei, L. Ding, Y. Cao, C. Zan, A. Yates, and D. Tao, "Unsupervised dense retrieval with relevance-aware contrastive pre-training," in *Findings of the Association for Computational Linguistics: ACL 2023*. Association for Computational Linguistics, 2023, p. 10932–10940. [Online]. Available: <http://dx.doi.org/10.18653/v1/2023.findings-acl.695>
- [14] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel *et al.*, "Retrieval-augmented generation for knowledge-intensive nlp tasks," *Advances in Neural Information Processing Systems*, vol. 33, pp. 9459–9474, 2020.
- [15] C.-Y. Lin, "Rouge: A package for automatic evaluation of summaries," *Proceedings of ACL Workshop*, pp. 74–81, 2004.
- [16] A. Mahalingam, V. K. Gande, A. Chadha, V. Jain, and D. Chaudhary, "Sketch: Structured knowledge enhanced text comprehension for holistic retrieval," *arXiv preprint arXiv:2412.15443*, 2024, workshop on Generative AI and Knowledge Graphs (GenAIK) at COLING 2025.
- [17] R. Nogueira and K. Cho, "Passage re-ranking with bert," 2020. [Online]. Available: <https://arxiv.org/abs/1901.04085>
- [18] K. Papineni, S. Roukos, T. Ward, and W.-J. Zhu, "Bleu: a method for automatic evaluation of machine translation," *Proceedings of ACL*, pp. 311–318, 2002.
- [19] "pgvector: Open-source vector similarity search for postgres," <https://github.com/pgvector/pgvector>, 2021, [Online]. Accessed: Oct. 15, 2025.
- [20] S. Robertson and H. Zaragoza, "The probabilistic relevance framework: Bm25 and beyond," *Foundations and Trends® in Information Retrieval*, vol. 3, no. 4, p. 333–389, 2009.
- [21] S. Sanniboina, S. Trivedi, and S. Vijayaraghavan, "Lore: Logit-ranked retriever ensemble for enhancing open-domain question answering," *arXiv preprint arXiv:2410.10042*, 2024, version 1, 13 Oct 2024.
- [22] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. u. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems*, I. Guyon, U. V. Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, and R. Garnett, Eds., vol. 30. Curran Associates, Inc., 2017.
- [23] J. Wei, X. Wang, D. Schuurmans, M. Bosma, B. Ichter, F. Xia, E. H. Chi, Q. V. Le, and D. Zhou, "Chain-of-thought prompting elicits reasoning in large language models," in *Proceedings of the 36th International Conference on Neural Information Processing Systems*, ser. NIPS '22. Red Hook, NY, USA: Curran Associates Inc., 2022.
- [24] T.-L. Yang, J.-S. Liu, Y.-H. Tseng, and J.-S. R. Jang, "Knowledge retrieval based on generative ai," 2025. [Online]. Available: <https://arxiv.org/abs/2501.04635>
- [25] S. Yao, J. Zhao, D. Yu, N. Du, I. Shafraan, K. Narasimhan, and Y. Cao, "React: Synergizing reasoning and acting in language models," in *11th International Conference on Learning Representations, ICLR 2023*, 2023.
- [26] K. Zhong, B. Suleiman, A. Erradi, and S. Chen, "Semrag: Semantic knowledge-augmented rag for improved question-answering," 2025. [Online]. Available: <https://arxiv.org/abs/2507.21110>